

Yilong Zhao (赵恽龙)

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RESEARCH INTEREST

DRAM-based processing-in-memory (PIM), including memory management, task scheduling, and application acceleration for PIM architectures.

EDUCATION

Sep. 2022 – Jun. 2026 (Expected) **Shanghai Jiao Tong University (SJTU), Shanghai, China**

Ph.D. Student, Major: Computer Science and Technology

Advisor: Prof. Li Jiang

Expected Graduation: Jun. 2026

Sep. 2018 – Mar. 2021 **Shanghai Jiao Tong University (SJTU), Shanghai, China**

M.Eng., Major: Computer Technology

Advisor: Prof. Li Jiang

Sep. 2014 – Jun. 2018 **Shanghai Jiao Tong University (SJTU), Shanghai, China**

B.Eng., Major: Electronic Science and Technology

PUBLICATIONS & PATENT (Full list can be found [here](#))

First Author:

- **Yilong Zhao**, Fangxin Liu, Onur Mutlu, Mingyu Gao, Jian Liu, Li Liang, and Haibing Guan, "COMET: A Cooperative Scheduling Framework for Concurrent PIM/CPU Execution on Mobile Devices", in *Proceedings of the 53rd International Symposium on Computer Architecture (ISCA'26, CCF-A, Accepted)*
- **Yilong Zhao**, Fangxin Liu, Zongwu Wang, Mingjian Li, Mingxing Zhang, Chixiao Chen, and Li Jiang, "BLADE: Boosting LLM Decoding's Communication Efficiency in DRAM-based PIM", in *Proceedings of the 31st Asia and South Pacific Design Automation Conference (ASP-DAC'26, CCF-C)*
- **Yilong Zhao**, Mingyu Gao, Huanchen Zhang, Fangxin Liu, Gongye Chen, He Xian, Haibing Guan, and Li Jiang, "PUSHtap: PIM-based In-Memory HTAP with Unified Data Storage Format, " in *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25, CCF-A)*
- **Yilong Zhao**, Fangxin Liu, Xiaoyao Liang, Mingyu Gao, Naifeng Jing, Chengyang Gu, Qidong Tang, Tao Yang, and Li Jiang, "STAMP: Accelerating Second-order DNN Training Via ReRAM-based Processing-in-Memory Architecture, " in *Proceedings of the 16th International Symposium on Advanced Parallel Processing Technology (APPT'25)*
- **Yilong Zhao**, Mingyu Gao, Fangxin Liu, Yiwei Hu, Zongwu Wang, Han Lin, Ji Li, He Xian, Hanlin Dong, Tao Yang, Naifeng Jing, Xiaoyao Liang, and Li Jiang, UM-PIM: DRAM-based PIM with Uniform & Shared Memory Space, in *Proceedings of the 51st International Symposium on Computer Architecture (ISCA'24, CCF-A)*
- Weidong Cao, **Yilong Zhao** (co-first author), Adith Bolor, Yinhe Han, Xuan Zhang, and Li Jiang, "Neural-PIM: Efficient Processing-In-Memory with Neural Approximation of Peripherals, " in *IEEE Transactions on Computers*, (TC, 2021, CCF-A)

- **Yilong Zhao**, Zhezhi He, Naifeng Jing, Xiaoyao Liang, and Li Jiang. Re2PIM: A Reconfigurable ReRAM-Based PIM Design for Variable-Sized Vector-Matrix Multiplication. In Proceedings of the 2021 on Great Lakes Symposium on VLSI (GLSVLSI'21, CCF-C)

Others:

- Yiwei Hu, Fangxin Liu, Zongwu Wang, **Yilong Zhao**, Tao Yang, Haibing Guan, and Li Jiang, "PLAIN: Leveraging High Internal Bandwidth in PIM for Accelerating Large Language Model Inference via Mixed-Precision Quantization", In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design, (ICCAD'25).
- Fangxin Liu, Wenbo Zhao, Zongwu Wang, **Yilong Zhao**, Tao Yang, Yiran Chen, Li Jiang, "IVQ: In-Memory Acceleration of DNN Inference Exploiting Varied Quantization", in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, (TCAD, 2022)
- Fangxin Liu, Wenbo Zhao, Zhezhi He, Zongwu Wang, **Yilong Zhao**, Yongbiao Chen, and Li Jiang, "Bit-Transformer: Transforming Bit-level Sparsity into Higher Performance in ReRAM-based Accelerator", In Proceedings of the 40th International Conference on Computer-Aided Design (ICCAD '21).
- Fangxin Liu, Wenbo Zhao, Zhezhi He, Zongwu Wang, **Yilong Zhao**, Tao Yang, Naifeng Jing, Xiaoyao Liang, and Li Jiang, "SME: ReRAM-based Sparse-Multiplication-Engine to Squeeze-Out Bit Sparsity of Neural Network," In Proceedings of the 39th IEEE International Conference on Computer Design (ICCD'21).
- Tao Yang, Dongyue Li, Yibo Han, **Yilong Zhao**, Fangxin Liu, Xiaoyao Liang, Zhezhi He, and Li Jiang, "PIMGCN: A ReRAM-Based PIM Design for Graph Convolutional Network Acceleration", In Proceedings of the 58th ACM/IEEE Design Automation Conference, (DAC'21).
- Ziqi Meng, Weikang Oian, **Yilong Zhao**, Yanan Sun, Rui Yang, and Li Jiang, "Digital Offset for RRAM-based Neuromorphic Computing: A Novel Solution to Conquer Cycle-to-cycle Variation," In Proceedings of the 24th Conference on Design, Automation and Test in Europe, (DATE'2021).
- Yanan Sun, Chang Ma, Zhi Li, **Yilong Zhao**, Jiachen Jiang, Weikang Qian, Rui Yang, Zhezhi He and Li Jiang, "Unary Coding and Variation-Aware Optimal Mapping Scheme for Reliable ReRAM-based Neuromorphic Computing," in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, (TCAD, 2021)
- Zhuoran Song, **Yilong Zhao**, Yanan Sun, Xiaoyao Liang and Li Jiang. "ESNreram: An Energy-Efficient Sparse Neural Network Based on Resistive Random-Access Memory, " In Proceedings of the 2020 on Great Lakes Symposium on VLSI, (GLSVLSI'20).
- Chaoqun Chu, Yanzhi Wang, **Yilong Zhao**, Xiaolong Ma, Shaokai Ye, Yunyan Hong, Xiaoyao Liang, Yinhe Han and Li Jiang. "PIM-Prune: Fine-Grain DCNN pruning for Crossbar-based Process-In-Memory architecture, " In Proceedings of the 57th ACM/IEEE Design Automation Conference, (DAC'20).

Patent

- Fangxin Liu, Li Jiang, **Yilong Zhao**, " Decoding acceleration methods, systems, devices, and readable storage media for large language models," Application No. 2025118583915.5
- Li Jiang, **Yilong Zhao**, "Reconfigurable Architecture, Accelerator, Circuit Deployment and Dataflow Methods," Authorization No. CN112181895B
- Li Jiang, **Yilong Zhao**, Xiaosong Cui, Yun Chen, Jianxing Liao, "Neural Network Circuit," Authorization No. CN114004344A

SCIENTIFIC RESEARCH EXPERIENCE

Shanghai Jiao Tong University, ACA-IMPACT Lab,

Shanghai Qizhi Institute

PIM Implementation Towards Optical Communication Project

Mar. 2021 – Jul. 2022

The purpose of the research is to realize the receiver of optical communication and wireless communication based on the integrated technology of storage and calculation. I am responsible for the following tasks:

- Design the overall architecture of optical communication and wireless communication systems based on the integration of storage and calculation, including operator splitting and algorithm reconstruction.
- Realize the circuit simulation of some operators.
- Aiming at the high-power calculation module to achieve a lower calculation amount than the existing numerical algorithm under the conditions of allowable error.

Shanghai Jiao Tong University, ACA Lab

ReRAM-based Efficient and Reliable DNN Accelerator Project

Apr. 2019 – Apr. 2020

The project investigates the enhancement of computational reliability and the utilizes sparsity to improve energy efficiency in ReRAM-based DNN accelerator. I am responsible for the following work:

- Design and code a cycle-accurate simulator for the ReRAM-based NN accelerator. The simulator is built based on GEM5.
- Rewrite the simulator to evaluate the reliability and performance of architecture for pruned NN, The results of the simulator are used as an important metric for the project evaluation.
- Design a ReRAM-based DNN accelerator for pruned NN.